

AccessEasy Combined Production Test Cases Suite (Flow-Wise & Tab-Wise)

This document contains a consolidated verification suite for the AccessEasy application. It includes: -

Section 1: End-to-End Core Business Flows: Deep-dives into critical cross-component user journeys (Auth, Scanners, Visitor Registrations, etc.). - **Section 2: Sidebar Tab-Wise Page Validations:** Quick validation checklists for all 13 pages linked in the side navigation bar.

Each test is mapped to its corresponding **UI Verification**, **API Response Validation**, and **Load & Performance Test Cases**.

SECTION 1: E2E Business Flows

■ Flow 1: Authentication & Session Security

■ UI Verification Cases

UI-1.1: Mobile OTP Form Validation

- **Action:** Load `/login`, type an alphabetical string in the mobile number field, then type `12345`.
- **Assertion:** Input field accepts numeric inputs only. Character limit is capped at 10-15 digits. If invalid, the validation message `"Please enter a valid mobile number"` triggers inline.

UI-1.2: Dark/Light Mode Inactivity Lock Overlay

- **Action:** Manually manipulate activity time to exceed 15 mins.
- **Assertion:**
 - A modal overlay is appended directly to the `<body>` element.
 - Backdrop is set to blurred (`backdrop-filter: blur(6px)`).
 - Background styling switches dynamically matching dark mode (`linear-gradient(145deg, #18181b, #09090b)`).
- All navigation buttons are covered and keyboard shortcuts are ignored.

■ API Response Validation Cases

API-1.1: Generate Phone OTP Response Assertions

- **Endpoint:** `POST ${VITE_KN_API_URL}/auth-service`
- **Request Payload:** `json { "action": "generate-otp", "phone": "9876543210", "userApp": "accesseasy" }`
- **Assertions:**

- Response status code must be **200 OK**.
- JSON Response schema assertion: `json { "success": true, "otp_session_uuid": "string (UUIDv4)" }`

API-1.2: Verify OTP Validation & Token Receipt

- **Endpoint:** `POST ${VITE_KN_API_URL}/auth-service`
- **Request Payload:** `json { "action": "verify-otp", "phone": "9876543210", "otp": "123456", "sessionUuid": "c2b64d0d-2ea3-4fae-a23c-f458e0a169b1" }`
- **Assertions:**
- Status code is **200 OK**.
- JSON Response must match the token profile: `json { "success": true, "token": "string (JWT)", "userData": { "id": "string/number", "first_name": "string", "last_name": "string", "email": "string", "role": { "name": "string" } } }`
- If invalid OTP, must return **400 Bad Request** or **200** with `success: false` and message `"OTP verification failed"`.

■ Load & Performance Test (k6 Script)

- **Goal:** Simulate peak hours logins where thousands of employees check in simultaneously.
- **Execution Target:** 200 virtual users (VUs) ramping up to 500 VUs over 2 minutes, maintaining load for 5 minutes.
- **k6 Simulation Script:** ````javascript import http from 'k6/http'; import { check, sleep } from 'k6';`

```
export const options = { stages: [ { duration: '1m', target: 200 }, // ramp up { duration: '3m', target: 200 }, // sustain { duration: '1m', target: 0 }, // cool down ], thresholds: { http_req_duration: ['p(95)<800'], // 95% of requests must complete under 800ms }, };
```

```
export default function () { const url = 'https://api-kn.fieldseasy.com/auth-service'; const payload = JSON.stringify({ action: 'generate-otp', phone: '919876543210', userApp: 'accesseasy' }); const params = { headers: { 'Content-Type': 'application/json' } }; const res = http.post(url, payload, params); check(res, { 'status is 200': (r) => r.status === 200, 'has session uuid': (r) => JSON.parse(r.body).otp_session_uuid !== undefined, }); sleep(1); } ```
```

■ Flow 2: Security & Authorization Key Scanner (Guard Portal)

■ UI Verification Cases

UI-2.1: Scanner Feedback Screen States

- **Action:** Trigger QR scanned responses.
- **Assertion:**
- **Success State:** Display a fullscreen emerald overlay (**bg-emerald-600**), verify employee photo, name, and access level render clearly. Display a physical checkmark animation.

- **Deactivated State:** Display a crimson overlay (`bg-rose-600`), verify warning text reads `"This token is invalid, expired, or holds insufficient clearance."`.

UI-2.2: Camera Access Error Message Layout

- **Action:** Block camera permissions in browser settings, then reload `/dashboard/authorize`.
- **Assertion:** UI switches container to black background, displays `<VideoOff />` SVG icon, and displays instructions to enable camera settings.

■ API Response Validation Cases

API-2.1: Fetch Valid QR Token Schema

- **Endpoint:** `GET ${VITE_API_URL}/items/qrgenerate?filter[qrcode][_eq]=...`
- **Assertions:**
- Status code is `200 OK`.
- JSON response contains necessary employee structure: `json { "data": [ { "id": "number", "qrcode": "string", "qrcode": true, "expires_at": "string (ISO8601)", "employeeId": { "id": "number", "firstName": "string", "lastName": "string", "dateOfLeaving": "string (date or null)", "access_level": { "status": "string (active | inactive)" } } } ] }`

API-2.2: Live Security Log Write Assertion

- **Endpoint:** `POST ${VITE_API_URL}/items/logs`
- **Request Payload:** `json { "tenant": "tenant-uuid", "employeeId": 12, "ValidLogs": "authorized", "action": "in", "mode": "throughApp", "date": "2026-06-18", "timeStamp": "22:22:30" }`
- **Assertions:**
- Status code is `201 Created` or `200 OK`.
- Response body contains the assigned log ID.

■ Load & Performance Test

- **Goal:** Simulate gate entries with continuous scans on shift changes.
- **k6 Simulation Script:** ````javascript import http from 'k6/http'; import { check, sleep } from 'k6';`

```
export const options = {
  vus: 50, // 50 concurrent security gates checking keys
  duration: '3m',
  thresholds: {
    http_req_duration: ['p(99)<1200'], // 99% of scans verified under 1.2s
  },
};
```

```
export default function () {
  const token = 'MOCK_BEARER_JWT_TOKEN';
  const params = {
    headers: {
      'Authorization': `Bearer ${token}`,
      'Content-Type': 'application/json'
    }
  };
}
```

```
// Simulate reading a QR token from gate
const url = 'https://api.fieldseasy.com/items/qrgenerate?filter[qrcode][_eq]=TEST_TOKEN_XYZ&fields=*,emp
```

```

const res = http.get(url, params);

check(res, {
  'status is 200': (r) => r.status === 200,
});

// Post security entry log
const logUrl = 'https://api.fieldseasy.com/items/logs';
const payload = JSON.stringify({
  tenant: 'tenant-1234',
  employeeId: 44,
  ValidLogs: 'authorized',
  action: 'in',
  mode: 'throughApp',
  date: '2026-06-18',
  timeStamp: '18:00:00'
});
const logRes = http.post(logUrl, payload, params);
check(logRes, {
  'log logged': (r) => r.status === 200 || r.status === 201
});

sleep(0.5); // Scan every 500ms

```

```

} ``

```

■ Flow 3: Branded Visitor Portals & Check-in

■ UI Verification Cases

UI-3.1: Form Fields Customizer Toggle Logic

- **Action:** Navigate to Portal Builder, disable "Photo Capture" toggle.
- **Assertion:**
- Photo upload section opacity drops to 60%.
- Nested "Required" toggle switches off and becomes disabled.
- Preview link updates dynamically.

UI-3.2: Mobile Check-in Form Validations

- **Action:** On public registration page `/visit/:id`, skip the Aadhar input field when configured as required.
- **Assertion:** On focus loss or submit click, input border displays `vp-input-error` outline (red color) and displays warning text "Government ID is required".

■ API Response Validation Cases

API-3.1: File Upload Handler Response (Selfies/ID proof)

- **Endpoint:** `POST ${VITE_API_URL}/files`
- **Request Format:** Multipart Form Data (`file` binary parameter)
- **Assertions:**
- Status code is `200 OK` or `201 Created`.
- JSON response must supply file ID: `json { "data": { "id": "uuid-string-of-uploaded-file", "filename_download": "string" } }`

API-3.2: Visitor Registration Knative Endpoint

- **Endpoint:** `POST ${VITE_KN_API_URL}/visitor-portal-flow/visitor`
- **Request Payload:** `json { "personName": "Tester Visitor", "mobileNumber": "9876543210", "email": "visitor@test.com", "startDate": "2026-06-18", "endDate": "2026-06-18", "startTime": "22:22:00", "endTime": "23:59:59", "status": "active", "quantity": 1, "tenant": { "tenantId": "t-100" }, "portalId": 5, "photo": "uuid-file-id" }`
- **Assertions:**
- Status code is `200 OK` or `201 Created`.
- Response must return generated QR string: `json { "status": "SUCCESS", "data": { "id": "number/string", "qrToken": "string (encrypted token or visitor json)" } }`

■ Load & Performance Test

- **Goal:** Simulate high volume visitor registrations during a corporate conference or event.
- **k6 Simulation Script:** ````javascript import http from 'k6/http'; import { check, sleep } from 'k6';`

```
export const options = { stages: [ { duration: '30s', target: 50 }, // ramp up { duration: '2m', target: 100 }, // sustained peak load { duration: '30s', target: 0 }, // ramp down ], thresholds: { http_req_failed: ['rate<0.01'], // less than 1% errors http_req_duration: ['p(95)<1500'], // 95% of registrations complete within 1.5s }, };
```

```
export default function () { const url = 'https://api-kn.fieldseasy.com/visitor-portal-flow/visitor'; const payload = JSON.stringify({ personName: LoadTest-Visitor-${__VU}-${__ITER}, mobileNumber: 90000000${__VU}, email: load-${__VU}@test.com, startDate: '2026-06-18', endDate: '2026-06-18', startTime: '09:00:00', endTime: '23:59:00', status: 'active', quantity: 1, tenant: { tenantId: 'tenant-99' }, portalId: 10 });
```

```
const params = { headers: { 'Content-Type': 'application/json' } };
const res = http.post(url, payload, params);

check(res, {
  'visitor created': (r) => r.status === 200 && JSON.parse(r.body).status === 'SUCCESS',
});

sleep(1);
```

```
} ``
```

■ Flow 4: Employee Credentials & NFC (My Access)

■ UI Verification Cases

UI-4.1: Render QR Key Display

- **Action:** Select "Generate New Key".
- **Assertion:**
- Spinner displays while fetching new token values.
- Image element updates with base64 encoded source (`data:image/png;base64...`).
- Key expiry countdown or status switches to "Active" display.

UI-4.2: RFID Management Card Revocation

- **Action:** Click the revocation ("X") button on an active RFID card indicator.
- **Assertion:**
- Standard alert confirmation pops up: "Are you sure you want to revoke this RFID card?".
- If cancelled, UI does not update. If confirmed, card layout updates back to input layout.

■ API Response Validation Cases

API-4.1: Retrieve User Profile (personalModule) details

- **Endpoint:** `GET ${VITE_API_URL}/items/personalModule?filter[assignedUser][_eq]=...`
- **Assertions:**
- Status code is `200 OK`.
- Schema contains card number and access level mappings: `json { "data": [ { "id": "number", "card_number": "string | null", "assignedAccessLevel": { "id": "number", "accessLevelName": "string" } } ] }`

API-4.2: Save Generated QR to directus db

- **Endpoint:** `POST ${VITE_API_URL}/items/qrgenerate`
- **Request Payload:** `json { "tenant": "tenant-id", "employeeId": 12, "accessLevelsId": 3, "qrcode": "AES-Encrypted-Key-String", "qrcode": true, "expires_at": "2026-06-19T22:22:30.000Z" }`
- **Assertions:**
- Status code is `200 OK` or `201 Created`.
- Body contains the created ID.

■ Load & Performance Test

- **Goal:** Measure key generation response time and database write limitations.
- **k6 Simulation Script:** ``javascript import http from 'k6/http'; import { check, sleep } from 'k6';

```
export const options = { vus: 80, duration: '2m', thresholds: { http_req_duration: ['p(95)<600'], // database write transaction must respond in under 600ms }, };
```

```
export default function () { const token = 'JWT MOCK EMPLOYEE KEY'; const params = { headers: { 'Authorization': 'Bearer ${token}', 'Content-Type': 'application/json' } };
```

```
const url = 'https://api.fieldseasy.com/items/qrgenerate';
const payload = JSON.stringify({
  tenant: 't-1234',
  employeeId: 44,
  accessLevelsId: 1,
  qrcode: 'AES256_STRESS_TEST_VALUE_XYZZY',
  qraccess: true,
  expires_at: '2026-06-19T18:00:00.000Z'
});

const res = http.post(url, payload, params);
check(res, {
  'qr stored': (r) => r.status === 200 || r.status === 201
});

sleep(1.5);
```

```
} ``
```

■ Flow 5: Core Master Settings Configuration (Branches)

■ UI Verification Cases

UI-5.1: Debounced Search UI State

- **Action:** Type 3 characters into search bar, verify no immediate visual table update, then wait.
- **Assertion:**
 - Table spinner shows up precisely 300ms after user pauses.
 - Search query matches the results matching typed parameters.

UI-5.2: Latitude/Longitude Floating Precision Check

- **Action:** Navigate to branches view list.
- **Assertion:** Coordinate chip displays numbers formatted with exactly four decimal places (e.g., **12.9716**), even if backend API delivers float variables with more decimals (e.g. **12.9715893**).

■ API Response Validation Cases

API-5.1: Get Branch Management Nodes

- **Endpoint:** `GET ${VITE_API_URL}/items/locationManagement`
- **Assertions:**
- Status code is `200 OK`.
- Body matches location schema: `json { "data": [ { "id": "number", "locType": "branch", "state": "string", "locdetail": { "locationName": "string", "address": "string", "pincodes": [ "string" ] }, "locmark": { "type": "Point", "coordinates": [ "number (longitude)", "number (latitude)" ] } } ] }`

■ Load & Performance Test

- **Goal:** Simulate administrative dashboard load querying branch listings under sustained search stress.
- **k6 Simulation Script:** ````javascript import http from 'k6/http'; import { check, sleep } from 'k6';`

```
export const options = { vus: 30, // 30 concurrent managers loading settings dashboard duration: '1m',  
thresholds: { http_req_duration: ['p(95)<400'], // branch list queries must resolve in < 400ms }, };
```

```
export default function () { const token = 'ADMIN_STRESS_JWT'; const params = { headers: {  
'Authorization': 'Bearer ${token}' } };
```

```
const url = 'https://api.fieldseasy.com/items/locationManagement?fields[]=orgLocation.orgName,locSize,lo  
const res = http.get(url, params);  
  
check(res, {  
  'branches list loaded': (r) => r.status === 200,  
});  
  
sleep(2);
```

```
} ```
```

SECTION 2: Sidebar Tab-Wise Page Validations

■ Tab 1: Dashboard / Live Insights (`/dashboard`)

■ UI Verification Cases

- **UI-T1.1 (Role Rendering):** Verify Guard login shows Security Overview metrics (Authorized, Denied, Auth Rate), Employee login shows Mobile Key and Attendance shortcuts, and Admins view all metric

cards.

- **UI-T1.2 (Live Activity Table):** Check that check-in events display in reverse chronological order. Status values must render green chips (**bg-emerald-50**) for authorized entries and red chips (**bg-rose-50**) for denials.
- **UI-T1.3 (Quick Action Prompts):** Verify that clicking "Register Door" opens the registration modal and "New Employee" triggers the onboarding popup.

■ API Response Validation Cases

- **API-T1.1 (Metric Queries):**
- **Endpoint:** **POST** `/accesseasy-dashboard-api/metrics`
- **Payload:** `{ "action": "main-dashboard", "tenantId": "t-100", "today": "2026-06-18", "role": "Admin" }`
- **Assertions:** Status **200 OK**. JSON must contain **doors** (integer), **employees** (integer), **authorized** (integer), **unauthorized** (integer), and **recentLogs** (array of objects).

■ Load & Performance Test Case

- **Load Profile:** Simulate 50 concurrent administrators loading and polling the live metric dashboard route every 5 seconds.
- **Target Metrics:** API must respond under **400ms** at 95th percentile (p95).

■ Tab 2: Visitor Dashboard (**/dashboard/visitors-overview**)

■ UI Verification Cases

- **UI-T2.1 (Overstay Warnings Alert):** Overstay list must render warnings in bold crimson font with immediate badge indicator if visit duration exceeds scheduled limit.
- **UI-T2.2 (Filter Panel):** Clicking filters (Today, This Week, Custom Date Range) must immediately partition table rows without page reload.

■ API Response Validation Cases

- **API-T2.1 (Get Visitor Metrics):**
- **Endpoint:** **GET** `/items/visitors?aggregate[count]=id&filter[status][_eq]=checked_in`
- **Assertions:** Status **200 OK**. Schema must return integer count matching current floor occupants.

■ Load & Performance Test Case

- **Load Profile:** Simulate 30 concurrent reception desk gates querying checked-in statuses during check-in rush hours (50 RPS).
- **Target Metrics:** DB lookup response time must resolve in under **500ms**.

■ Tab 3: Doors (/dashboard/access-control/doors)

■ UI Verification Cases

- **UI-T3.1 (Registration Form Validation):** Leaving Door Name blank on the "Register Door" form must block submission and highlight the input border in red with a "Door Name is required" label.
- **UI-T3.2 (Lock Status Display):** Locked doors must render a secure lock icon; open doors must flash a warning indicator.

■ API Response Validation Cases

- **API-T3.1 (Fetch Doors List):**
- **Endpoint:** `GET /items/doors?filter[tenant][tenantId][_eq]=...`
- **Assertions:** Status `200 OK`. Checks if door array contains `id`, `doorName`, `macAddress`, `status` (active/inactive).
- **API-T3.2 (Add Door):**
- **Endpoint:** `POST /items/doors`
- **Payload:** `{ "doorName": "Front Gate", "macAddress": "00:1A:2B:3C:4D:5E", "tenant": "t-100" }`
- **Assertions:** Status `201 Created` with created record JSON confirmation.

■ Load & Performance Test Case

- **Load Profile:** Simulate a bulk import upload where 100 doors are provisioned simultaneously via CSV upload parser.
- **Target Metrics:** Maximum queue write duration must not exceed **2.5s**.

■ Tab 4: Groups (Access Clearance)

(/dashboard/easy-access/configurators/access-levels)

■ UI Verification Cases

- **UI-T4.1 (Checkbox Selector Grid):** Checkbox matrix showing doors vs schedules must render correctly. Select All / Deselect All headers must instantly toggle corresponding items.
- **UI-T4.2 (Revoke Flag Visuals):** Inactivating an access group must update status labels to red badge.

■ API Response Validation Cases

- **API-T4.1 (Query Access clearance levels):**

- **Endpoint:** `GET /items/accesslevels?fields=id,accessLevelName,status`
- **Assertions:** Status `200 OK`. Verify inactive groups are correctly flagged.

■ Load & Performance Test Case

- **Load Profile:** Simulate 40 bulk revisions where security admins adjust clearance permissions across multiple access groups.
 - **Target Metrics:** Average request duration under **800ms**.
-

■ Tab 5: Guards (`/dashboard/guards`)

■ UI Verification Cases

- **UI-T5.1 (Add Guard Input checks):** Mobile field must validate character count. Email field must check for @ character validation.
- **UI-T5.2 (Gate Assignment Select Menu):** Changing location assignment must immediately change the mapped branch value displayed in the table row.

■ API Response Validation Cases

- **API-T5.1 (Create Guard Login):**
- **Endpoint:** `POST /items/guards`
- **Payload:** `{ "firstName": "Ramesh", "lastName": "Kumar", "mobile": "9988776655", "branchId": 5 }`
- **Assertions:** Status `201 Created`. JSON response returns the new user profile object.

■ Load & Performance Test Case

- **Load Profile:** Simulate bulk check assignment query during change of security shifts (100 simultaneous requests).
 - **Target Metrics:** Read endpoints response latency below **350ms**.
-

■ Tab 6: Employees (`/dashboard/easy-access/employees`)

■ UI Verification Cases

- **UI-T6.1 (Deactivation Date Picker):** Choosing a resignation date from the calendar must display warning: `"Deactivating account locks out digital keys."`
- **UI-T6.2 (Search & Filter Column):** Search bar must execute matching queries. Column filters (e.g. Filter by Department) must instantly update rows.

■ API Response Validation Cases

- **API-T6.1 (Deactivate Employee):**
- **Endpoint:** `PATCH /items/personalModule/:id`
- **Payload:** `{ "dateOfLeaving": "2026-06-17" }`
- **Assertions:** Status `200 OK`. Checked-out state verified on database schema.

■ Load & Performance Test Case

- **Load Profile:** Run bulk query simulation parsing 1,000 employee profile data cards with nested access level records.
- **Target Metrics:** Response throughput $\geq$ **300 requests/sec** with p99 $<$ **1.5s**.

■ Tab 7: Visitor Portals (`/dashboard/visitor-portals` & `/visit/:id`)

■ UI Verification Cases

- **UI-T7.1 (Mobile Camera Selfie Feed):** Click capture photo field. Camera feed must open. Swapping facing camera front/rear on mobile must display live mirroring correctly.
- **UI-T7.2 (Fields Customizer Sync):** Hide Government ID option in builder. Verify visitor check-in form hides the ID input field.

■ API Response Validation Cases

- **API-T7.1 (Document Proof upload):**
- **Endpoint:** `POST /files`
- **Format:** Multipart form binary upload.
- **Assertions:** Status `200 OK`. Must yield valid UUID token string identifier.

■ Load & Performance Test Case

- **Load Profile:** Run k6 script loading 120 visitor registrations checking in simultaneously, upload image binaries (100KB each).
- **Target Metrics:** System must resolve visitor registrations under **1.8s** with no database deadlock errors.

■ Tab 8: Scheduled Reports (`/dashboard/report-automation`)

■ UI Verification Cases

- **UI-T8.1 (Cron Expression builder):** Selecting "Every Monday" must set default input cron values to `0 9 * * 1`. Custom input overrides must validate expression before saving.

- **UI-T8.2 (E-mail Address tags):** Adding target recipients must render clean chip elements with close ('x') action handles.

■ API Response Validation Cases

- **API-T8.1 (Save Automation Task):**
- **Endpoint:** `POST /items/reportAutomation`
- **Payload:** `{ "reportName": "Weekly Activity", "cron": "0 9 * * 1", "format": "pdf", "emails": ["manager@co.com"] }`
- **Assertions:** Status `201 Created`.

■ Load & Performance Test Case

- **Load Profile:** Trigger 50 manual immediate PDF document builds representing 10,000 logs records.
- **Target Metrics:** Backend processing worker threads must complete compilation tasks without leaking system memory.

■■ Tab 9: Settings - Devices (`/dashboard/settings/devices`)

■■ UI Verification Cases

- **UI-T9.1 (Online Pulse indicator):** Active devices must render a pulsing green dot, offline devices render solid grey status dots.
- **UI-T9.2 (Sync Logs Panel):** Terminal command consoles must display logs line-by-line with auto-scroll lock capability.

■ API Response Validation Cases

- **API-T9.1 (Device Health Status request):**
- **Endpoint:** `GET /items/devicesManager?fields=id,deviceName,isOnline,lastSeen`
- **Assertions:** Status `200 OK`. Verify timestamp structure.

■ Load & Performance Test Case

- **Load Profile:** Simulate 500 devices concurrently checking-in (sending status heartbeats) every 30 seconds.
- **Target Metrics:** CPU utilization must remain below **40%** at server gateway node.

■ Tab 10: Settings - Event Logs (`/dashboard/settings/logs`)

■ UI Verification Cases

- **UI-T10.1 (Lazy Load Infinite Scroll):** Scrolling to the bottom of the table logs must load the next chunk of logs seamlessly.
- **UI-T10.2 (CSV/Excel Export Button):** Verify file download dialog opens instantly and names the file dynamically.

■ API Response Validation Cases

- **API-T10.1 (Filtered Logs Fetch):**
- **Endpoint:** `GET /items/logs?filter[ValidLogs][_eq]=authorized&limit=50`
- **Assertions:** Status `200 OK`. Verify schema returns employee names and check-in times.

■ Load & Performance Test Case

- **Load Profile:** Run continuous read queries pulling 5,000 log events with multiple filters (Branch, Date Range, Status).
- **Target Metrics:** Latency below **600ms**.

■ Tab 11: My Access (`/dashboard/my-access`)

■ UI Verification Cases

- **UI-T11.1 (Visual QR code generation):** Generating key must display canvas QR pass.
- **UI-T11.2 (RFID Register Option):** Verify entering a valid RFID tag updates status to Active.

■ API Response Validation Cases

- **API-T11.1 (Register Token):**
- **Endpoint:** `POST /items/qrgenerate`
- **Assertions:** Status `201 Created`. Token expires in exactly 24 hours.

■ Load & Performance Test Case

- **Load Profile:** Simulate 200 employees generating keys at the start of shift.
- **Target Metrics:** Response time < **700ms**.

■ Tab 12: My Attendance (`/dashboard/my-attendance`)

■ UI Verification Cases

- **UI-T12.1 (Calendar Timeline):** Checked-in days must display green, absent days display red.
- **UI-T12.2 (Regularisation Form):** Clicking "Request Change" must display form validator modal.

■ API Response Validation Cases

- **API-T12.1 (Get Attendance Logs):**
- **Endpoint:** `GET /items/personalModule/attendance`
- **Assertions:** Status `200 OK`. Returns attendance records.

■ Load & Performance Test Case

- **Load Profile:** Simulate 300 employees checking calendars simultaneously.
- **Target Metrics:** Response time < **500ms**.

■ Tab 13: Authorize (`/dashboard/authorize`)

■ UI Verification Cases

- **UI-T13.1 (Laser Animation scan):** Verify camera feed displays a red animated laser line overlay.
- **UI-T13.2 (Visual Result Screens):** Valid QR results render green screens, invalid QRs render red screens.

■ API Response Validation Cases

- **API-T13.1 (Verify Token API):**
- **Endpoint:** `GET /items/qrgenerate?filter[qrcode][_eq]=...`
- **Assertions:** Status `200 OK`. Must query status, expiration, and access restrictions.

■ Load & Performance Test Case

- **Load Profile:** Run continuous checks simulating 100 gates scan requests.
- **Target Metrics:** Average request-response time must resolve within **1.0s**.